Information Technology

Class XI — Condensed Lecture Notes
Computer Systems · Computational Thinking · Python · Databases · Emerging Tech · Digital Literacy

Draft · SVG diagrams · MathJax · Flexbox layout · Aligned to uploaded curriculum PDFs

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Unit 3 — Programming in Python

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Unit 1 — Computer Systems

Learning outcomes (summary)

- Differentiate analog vs digital representation; build and simulate logic circuits.
- Apply Boolean algebra, truth tables, and Logisim for gate-level design.
- Explain SDLC phases; compare Waterfall and Agile (Scrum) with case studies.
- Describe OSI and TCP/IP layers, encapsulation, and common protocols.

1.1 Digital foundations

Discrete quantities are countable (students in a class, books on a shelf). **Continuous** quantities vary smoothly (temperature, time, height). Digital systems encode information with two states: bit `0` (LOW/OFF/FALSE) and bit `1` (HIGH/ON/TRUE).

Analog signals can take many amplitude levels; **digital signals** switch between defined levels (typically two). Digital storage and copying resist noise better than analog chains of amplification.

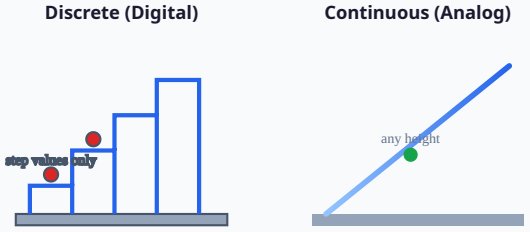


Figure 1.1 — Stairs (discrete) vs ramp (continuous)

1.2 Boolean algebra & logic gates

Boolean variables $A, B, C \in \{0, 1\}$. Basic operations:

- AND (product): $A \cdot B$ — true only when both inputs are 1.
- OR (sum): $A + B$ — true when at least one input is 1.
- NOT (complement): $\overline{A}$.

De Morgan's laws (exam favourites):

$$\overline{A \cdot B} = \overline{A} + \overline{B}, \qquad \overline{A + B} = \overline{A} \cdot \overline{B}$$

Any Boolean function can be written in **SOP** (sum of products) or **POS** (product of sums) form, e.g. $F = A\overline{B} + BC$.



Figure 1.2 — Standard logic gates

Logisim workflow

1. Create project → place gates from Explorer.
2. Add input pins (square) and output pin/LED (circle).
3. Wire inputs → gate → output; toggle inputs to verify truth table.
4. Example: 2-input AND with $Y = A \cdot B$.

Sample truth table — 2-input XOR

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

1.3 Software Development Life Cycle (SDLC)

SDLC phases: **planning / requirements** → **analysis** → **design** → **implementation (coding)** → **testing** → **deployment** → **maintenance**. Choosing a model depends on requirement stability, stakeholder count, and need for early feedback.



Figure 1.3 — Waterfall vs Agile (Scrum sprints)

Waterfall suits fixed specs (e.g. library management system with clear modules: catalog, issue/return, fines).

Agile / Scrum suits evolving mobile apps (e.g. school portal with shifting features): 2-week sprints, product backlog, review with users each increment.

1.4 Networking — OSI & TCP/IP



Figure 1.4 — OSI reference model

Encapsulation: each layer adds its header (and sometimes trailer) as data moves down the stack; the receiver strips headers moving up.

TCP/IP (practical Internet stack): Application → Transport (TCP/UDP) →

Internet (IP) → Network Access (link + physical). OSI is a teaching model; TCP/IP is what most implementations follow.

Layer (OSI)	PDU / role	Examples
Application (7)	Data	HTTP, DNS, SMTP
Transport (4)	Segment	TCP (reliable), UDP (fast)
Network (3)	Packet	IP, routers
Data Link (2)	Frame	MAC, switches
Physical (1)	Bits	Cables, Wi-Fi radio

Unit 2 — Computational Thinking

Learning outcomes (summary)

- Decompose problems; recognize patterns; abstract details; design algorithms.
- Write pseudocode; state properties of good algorithms (correct, finite, efficient).
- Apply search (linear, binary) and sorting (bubble, selection, insertion) algorithms.

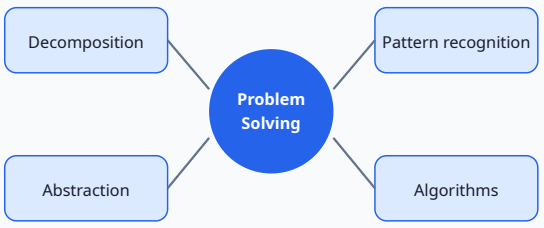


Figure 2.1 — Four pillars of computational thinking

Algorithm — finite, step-by-step procedure solving a class of problems. **Pseudocode** mixes English with structured keywords (`IF`, `WHILE`, `FOR`) without language syntax.

> **Good algorithm properties:** input/output defined, steps are clear, must terminate, each step is feasible, correctness for all valid inputs.

2.1 Searching

- **Linear search** — scan all n elements; worst case $O(n)$.
- **Binary search** — requires sorted array; halve search space each step; $O(\log n)$.

2.2 Sorting (know any two)

Algorithm	Idea	Typical use
Bubble sort	Swap adjacent out-of-order pairs until sorted	Teaching; small n
Selection sort	Repeatedly pick minimum for current position	Simple, few swaps
Insertion sort	Build sorted prefix by inserting each element	Nearly sorted data

Unit 3 — Programming in Python

Learning outcomes (summary)

- Distinguish low-level vs high-level languages; use Python IDEs (e.g. VS Code).
- Apply variables, data types, operators, control structures, functions, and modules.
- Understand frameworks/libraries ecosystem for career growth.

3.1 Language & environment

Python is **interpreted**, **high-level**, and emphasizes readable syntax. Use virtual environments and linters in VS Code; run via integrated terminal.

3.2 Control structures

- **Sequence** — statements execute in order.
- **Selection** — `if / elif / else` for branching.
- **Iteration** — `while` and `for` loops (often with `range()`).

```
# Selection + loop example
total = 0
for mark in [72, 85, 91]:
    if mark ≥ 50:
        total += mark
print("Passing sum:", total)
```

3.3 Functions & good style

Functions encapsulate reusable logic; use descriptive names, docstrings, and avoid magic numbers. Import standard modules (`math` , `random`) before third-party packages.

Unit 4 — Data and Analysis

Learning outcomes (summary)

- Relate data, information, database, and DBMS; identify components and keys.
- Design tables, relationships, forms, queries, and reports (e.g. LibreOffice Base).
- Apply integrity constraints and appropriate data types.

4.1 Core concepts

Data are raw facts; **information** is processed, meaningful data. A **database** is an organized collection; a **DBMS** manages storage, security, and concurrent access.

4.2 Keys & integrity

- **Primary key** — uniquely identifies a row.
- **Foreign key** — links to a primary key in another table (referential integrity).
- **Constraints** — NOT NULL, UNIQUE, CHECK, DEFAULT.

4.3 Forms & user interface

Forms provide structured data entry: identify data source, layout fields with validation, use wizards for quick prototypes, then refine labels and tab order for usability.

Unit 5 — Emerging Technologies

Learning outcomes (summary)

- Define AI, IoT, and data analytics at a high level.
- Discuss benefits, risks, ethics, and assistive applications.

5.1 Artificial Intelligence

AI systems perceive, reason, learn, and act. Machine learning is a subset where models improve from data rather than explicit rule lists only.

5.2 Internet of Things (IoT)

Network of sensors/actuators exchanging data with cloud services — smart home, agriculture, healthcare monitors. Security (firmware updates, strong passwords) is critical.

5.3 Data analytics pipeline

Collect → clean → explore → model → visualize → decide. Emphasize privacy, consent, and bias when interpreting results.

Unit 6 — Digital Literacy

Learning outcomes (summary)

- Compare qualitative vs quantitative data; choose collection strategies.
- Practice safe, ethical use of digital tools and online information.
- Plan small digital projects from problem statement to prototype.

6.1 Types of data

Type	Nature	Examples
Qualitative	Descriptive, categories	Interview themes, open-ended survey text
Quantitative	Numeric, measurable	Test scores, response times, counts

6.2 Collection methods

- **Interviews / surveys** — attitudes and self-reported behaviour.
- **Observation** — record actions in context.
- **Prototypes & simulations** — test ideas before full build.

Digital literacy includes evaluating sources (CRAAP: Currency, Relevance, Authority, Accuracy, Purpose), managing digital footprint, and respecting copyright.

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Class XII — Condensed Lecture Notes
HCI · Computational Thinking · Data Structures · Data Analysis · AI/ML · Entrepreneurship

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Unit 1 — Human–Computer Interaction

Learning outcomes (summary)

- Explain HCI fundamentals, UX/UI principles, and accessibility.
- Analyse user needs; wireframe and prototype interfaces (e.g. Figma).
- Test usability: efficiency, learnability, error rate, satisfaction.

1.1 Interaction paradigms

Traditional interaction — keyboard, mouse, windows/icons/menus/pointers (WIMP). **Natural interaction** — touch, voice, gesture, pen; reduces visible chrome but needs clear affordances.

HCI goal: systems that are **useful, usable, and satisfying** for target users in real contexts (school app vs public kiosk differ).

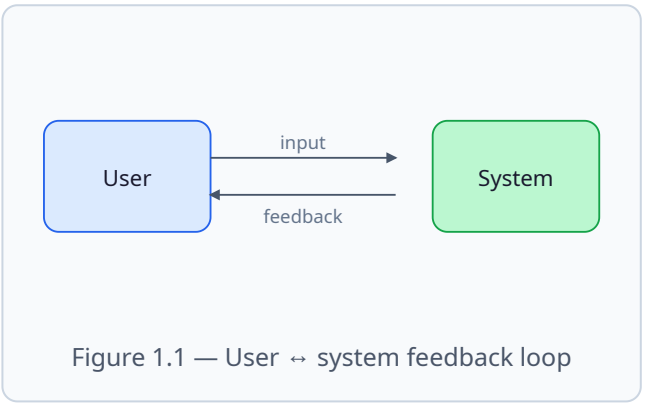


Figure 1.1 — User ↔ system feedback loop

1.2 Accessibility & inclusive design

- Sufficient colour contrast; do not rely on colour alone.
- Readable typography; scalable text; colour-blind-safe palettes.
- Keyboard navigation and screen-reader labels for interactive controls.

1.3 Design process (Figma)

1. Personas & scenarios from stakeholder interviews.
2. Low-fidelity wireframes → interactive prototype.
3. Heuristic evaluation + user testing; iterate on findings.

Unit 2 — Computational Thinking (Advanced)

Learning outcomes (summary)

- Apply stepwise reasoning and invariants to harder problems.
- Analyse algorithm efficiency with Big-O notation.
- Select appropriate strategies (divide-and-conquer, greedy where suitable).

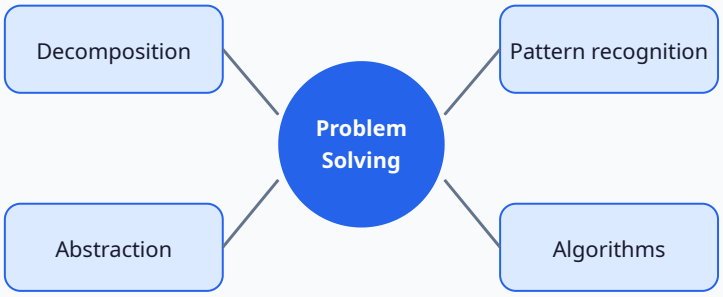


Figure 2.1 — Problem-solving framework

2.1 Complexity sketch

If an algorithm performs $3n^2 + 5n + 2$ operations, its growth is written $O(n^2)$. Compare:

$$O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2)$$

2.2 Stepwise reasoning

Break invariants into preconditions, loop guard, and postconditions — e.g. binary search maintains “answer lies in index range [low, high]”.

Unit 3 — Data Structures in Python

> **Learning outcomes (summary)**
>
> - Use lists, tuples, sets, dictionaries; choose structure by access pattern.
> - Implement stacks/queues conceptually; traverse nested structures.

3.1 Lists

Ordered, mutable sequences — indexing $O(1)$, append amortized $O(1)$, insert/delete in middle $O(n)$.

```
scores = [88, 76, 91]
scores.append(84)
scores.sort()
```

3.2 Dictionaries & sets

dict — key→value maps for fast lookup by key. **set** — unique elements; union/intersection for membership tests.

Structure	Ordered?	Mutable?	Typical use
list	Yes	Yes	Sequences, stacks (via append/pop)
tuple	Yes	No	Fixed records, dict keys
set	No	Yes	Unique tags, fast membership
dict	Insertion-ordered (Py3.7+)	Yes	Lookup tables, JSON-like data

Unit 4 — Data Analysis with Python

Learning outcomes (summary)

- Connect to SQLite; run CRUD with Python `sqlite3`.
- Load and analyse tabular data with pandas DataFrames.
- Interpret basic aggregates, filtering, and simple visualizations.

4.1 Database connectivity

Components: driver/API, connection string, cursor, commit/rollback. SQLite stores a whole database in one file — ideal for classroom projects.

```
import sqlite3
conn = sqlite3.connect("school.db")
cur = conn.cursor()
cur.execute("CREATE TABLE IF NOT EXISTS student (id INTEGER PRIMARY KEY, name TEXT)")
conn.commit()
conn.close()
```

4.2 pandas DataFrames

```
import pandas as pd
df = pd.read_csv("marks.csv")
print(df.describe())
passed = df[df["score"] >= 50]
```

Cleaning steps: handle missing values, normalize column names, validate types before plotting.

Unit 5 — Machine Learning, Neural Networks & Deep Learning

Learning outcomes (summary)

- Distinguish ML, neural networks, and deep learning.
- Name components: neurons, layers, weights, activation, loss, optimizer.
- Discuss training data, overfitting, and ethical deployment.

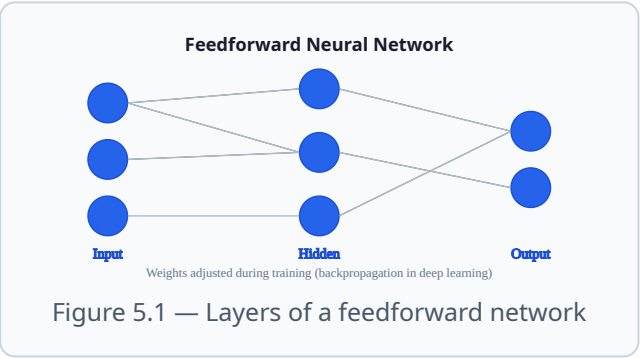


Figure 5.1 — Layers of a feedforward network

Machine learning — learn parameters from examples (supervised, unsupervised, reinforcement).

Neural network — connected units with non-linear activations; universal approximator in theory.

Deep learning — many hidden layers; excels at vision, speech, NLP when data and compute are sufficient.

Training minimizes loss $L(\theta)$ over weights θ using gradient descent variants:

$$\theta \leftarrow \theta - \eta \nabla_\theta L(\theta)$$

Exam checklist: training vs inference; labelled data; validation split; bias/variance; never deploy without testing on representative users.

Unit 6 — Entrepreneurship

Learning outcomes (summary)

- Differentiate entrepreneur vs entrepreneurship; spot opportunities.
- Move from problem → idea → prototype → pitch.
- Compare prototype fidelity: low / mid / high.

6.1 From problem to business idea

Validate pain points through interviews; define MVP scope; map stakeholders (customers, partners, regulators).

6.2 Prototype types

Fidelity	Form	Purpose
Low	Paper sketches, storyboards	Explore many concepts cheaply
Mid	Clickable wireframes (Figma)	Test flows with users
High	Working subset with real data	Pilot deployment, investor demo

6.3 Business model canvas (one slide)

Value proposition	Customer segments	Channels	Revenue & costs